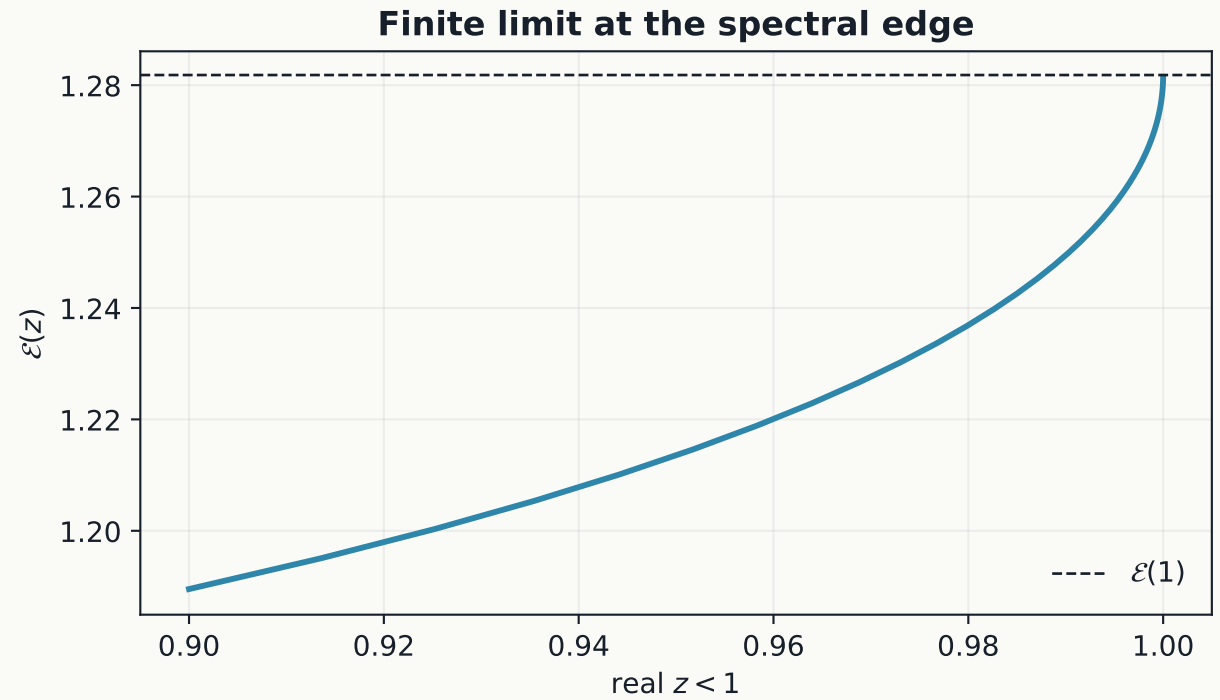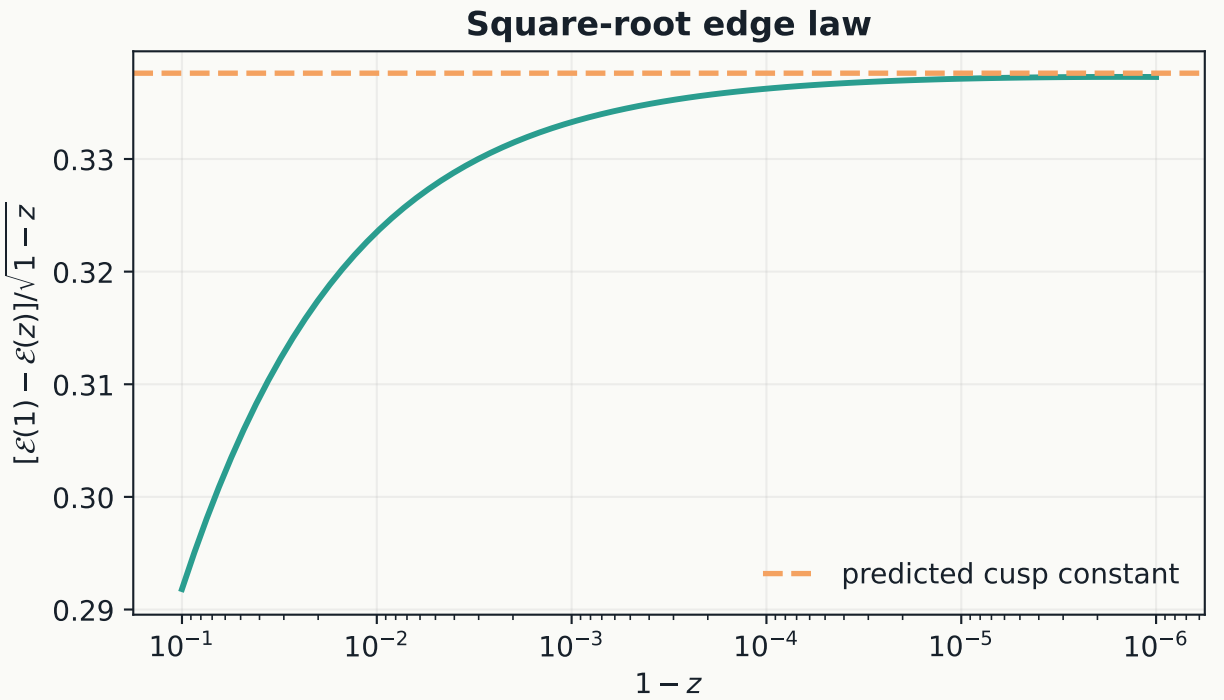The principal energy resolvent has a computable square-root cusp at its cut endpoint